

CoRoT-2b

R-0174

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_200816 · created 2026-07-16T18:10:40+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459077.8435 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.213 +/- 0.065
Transit depth (Rp/Rs)^2	4.53 +/- 2.77 [%]
Orbital Inclination (inc)	89.24 +/- 2.06
Airmass coefficient 1 (a1)	2.85 +/- 1.31
Airmass coefficient 2 (a2)	-0.323 +/- 0.093
Scatter in the residuals of the lightcurve fit is	34.61 %
Best Comparison Star	#1 - [538, 304]
Optimal Aperture	9.72
Optimal Annulus	15.14
Transit Duration (day)	0.0516 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.213 ± 0.065 vs 0.1667 (deviation 0.71σ)
Duration fit vs published	0.0516 d vs 0.095 d (deviation 10.09σ)
Mid-time O–C	6.3 min
Depth SNR	3.3
Residual scatter	34.61 %

Light curve

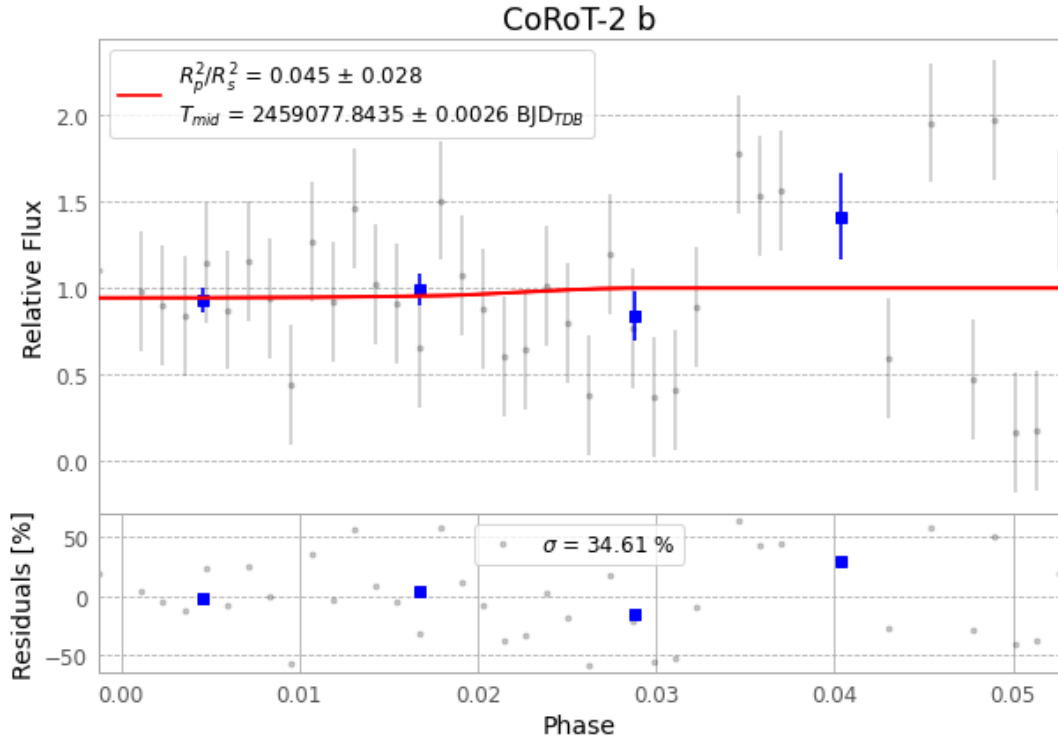


Figure 1 — FinalLightCurve_CoRoT-2 b_2020-08-16.png (SHA-256 c5390343bfa97609...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 142], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c93dd9781d5fae92...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9b8a76a

Data provenance

45 FITS frames (30 MB), CoRoT-2200816080014.FITS → CoRoT-2200816101814.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-200816.log (f5edffba13a5...), FinalLightCurve_CoRoT-2 b_2020-08-16.png (c5390343bfa9...), FinalLightCurve_CoRoT-2 b_2020-08-16.csv (06390137ddd6...)

Compute resources

Wall time 165.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 18:10Z.